

Biotechnology Tasks & Instructions — CodeAlpha

Biotechnology — Internship Overview

This internship program offers hands-on experience in biotechnology concepts and applications. CodeAlpha is a leading software development and research-driven company dedicated to fostering innovation in science and technology. The internship equips students with skills to analyze biological data, understand genetic engineering techniques, explore bioinformatics tools, and study industrial and medical biotechnology applications. Interns will gain practical exposure through research-based projects, case studies and simulations. Expert mentorship and a collaborative environment will prepare interns to contribute effectively to advancements in healthcare, agriculture and environmental sustainability.

Internship Perks

- Internship Offer Letter
 - Completion Certificate (QR Verified)
 - Unique ID Certificate
 - Letter of Recommendation (based on performance)
 - Job Opportunities / Placement Support
 - Resume Building Support
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Instructions for Interns

1. Share your internship status on **LinkedIn**, tagging **@CodeAlpha**.
 2. Complete the **assigned projects** within the mentioned time frame.
 3. Upload your complete file to **Google drive**:
`CodeAlpha_ProjectName`
 4. Post a **video explanation** of your project on LinkedIn.
 5. Submit your completed task using the **Submission Form**.
 6. **Complete any 2 or 3 out of the 4 tasks** listed below (from your domain).
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Biotechnology Task List

(Complete any 2 or 3 of the following tasks)

TASK 1: Research Report

Prepare a detailed report (1500–2000 words) on **Recent Advancements in Biotechnology** in one of the following areas:

- Genetic Engineering (CRISPR, Gene Therapy)
 - Industrial Biotechnology (Biofuels, Enzymes, Bioplastics)
 - Agricultural Biotechnology (GM crops, Biofertilizers)
 - Medical Biotechnology (Vaccines, Diagnostics, Stem Cell Therapy)
- 👉 Submit in PDF format with proper references (APA/MLA).

✓ TASK 2: Case Study Analysis

Choose one **real-world biotechnology startup/company** (e.g., Biocon, Moderna, Ginkgo Bioworks).

- Explain their biotechnology innovation/product.
 - Discuss applications, challenges, and market impact.
 - Prepare a **presentation (10–12 slides)**.
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✓ TASK 3: Laboratory/Practical Task (Virtual/Research-Based)

Since you may not have a physical lab, perform a **virtual simulation or research-based analysis**:

- DNA Extraction (from fruits/vegetables – home experiment) OR
 - Enzyme Activity Analysis (e.g., amylase, catalase) OR
 - Microbial Culture Research (case study on E. coli, yeast, or probiotics)
 - 👉 Document your **method, results, and conclusion** (with images/screenshots if applicable).
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✓ TASK 4: Mini-Project / Proposal

Prepare a **mini project proposal (5–6 pages)** on one of the following:

- Development of a low-cost biosensor for detecting diseases.
 - Use of biotechnology in sustainable agriculture.
 - AI/ML applications in biotechnology research.
 - Role of biotechnology in climate change solutions.
 - 👉 The proposal should include **Introduction, Problem Statement, Objectives, Methodology, Expected Outcome and References**.
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Important Note

Internship Completion Criteria:

To be eligible for the internship certificate, participants must complete a minimum of **two or three tasks**. Submitting only one task will be considered **incomplete** and certificates will **not** be issued in such cases.



Submission Details

A submission form will be shared in your respective **WhatsApp group**. You are required to submit your completed task only through that form. Please follow the instructions mentioned in the form carefully to ensure your submission is accepted.



Contact Information

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